ALIYAN FAISAL

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cuelara.com · invoiceinspect.app

RESEARCH PROFILE

Software and AI Systems Developer with a background in Information Technology (CGPA 3.80/4.00) specializing in applied AI systems, LLM orchestration, and Retrieval-Augmented Generation (RAG) architectures. Experienced in building scalable data pipelines, prompt engineering toolkits, and intelligent decision-support platforms. Strong foundation in computer science fundamentals, backend system design, and API-driven automation. Seeking admission into a research-oriented Master's program in Artificial Intelligence or Machine Learning to transition from applied AI systems engineering to foundational Machine Learning research.

RESEARCH INTERESTS

- **Applied LLMs & Prompt Engineering:** Context-window optimization, token reduction, and structured output evaluation.
- **Retrieval-Augmented Generation (RAG):** Document parsing, vector embeddings, and retrieval for grounded, cited answers.
- **LLM Reliability & Evaluation:** Cross-checking independent readings, deterministic verification, confidence and evidence, and evaluation harnesses.
- **Data-Centric Automation & Systems:** Scalable backend infrastructure for machine learning services and decision platforms.

EDUCATION

Karakoram International University Gilgit, Pakistan
Bachelor of Science in Information Technology **CGPA: 3.80 / 4.00** (2019 – 2023)

- **Scholarship:** HEC Ehsaas Scholarship (fully funded for the entire degree).
- **Core Coursework:** Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Database Systems, Operating Systems, Computer Networks, Distributed Systems, Software Engineering.
- **Academic Focus:** Applied computing, system architecture, database design, and algorithmic problem-solving.

TECHNICAL SKILLS

- **AI Engineering & RAG:** LLM APIs (OpenAI, Gemini, Anthropic, Groq, DeepSeek), Local LLMs, Tool and Function Calling, Retrieval-Augmented Generation (RAG), Vector Embeddings, Prompt Optimization, Context Window Management, Token Reduction.
- **Languages & Backend:** Python, PHP (Laravel), Node.js, Flask, TypeScript, JavaScript, RESTful APIs, SQL.
- **Data & Infrastructure:** Pandas, NumPy, PostgreSQL, MySQL, MongoDB, Vector Databases, Docker, Git, Linux Administration.

FEATURED AI & SOFTWARE PROJECTS

Cuelara – LLM Prompt Optimization & Context Engineering Suite cuelara.com
Creator & Lead Systems Engineer **2025 – Present**

- Architected a web-based toolkit featuring a **Prompt Optimizer**, **Token Reducer**, **Context Extractor**, and **Prompt Debugger** to streamline LLM inference pipelines.
- Engineered token-reduction algorithms and structural formatting rules to minimize context-window overhead and API latency while improving response consistency.
- Implemented context extraction pipelines to isolate system instructions from input noise prior to API submission.
- Shipped the toolkit through a web app, browser extension, public API and MCP server, with rate limiting based on the plan each user has purchased.

InvoiceInspect – Multimodal Invoice Verification & Error Detection System invoiceinspect.app
Creator & Lead Systems Engineer **2025 – Present**

- Built a system that finds arithmetic and consistency errors in PDF invoices (line totals, subtotals, tax, grand total, duplicate charges, missing fields) and reports each with evidence: expected value, printed value and page.
- Designed dual-path extraction, combining native text and OCR (Tesseract) with a vision model, cross-checked field by field; disagreements are marked "could not verify" instead of guessed.
- Kept verification deterministic: a rules engine decides, and the LLM only explains findings, with a provider-agnostic AI layer and fallback between providers.
- Built an evaluation harness (dataset generator and scorer); currently ∼**92%** end-to-end error-detection accuracy, with ongoing improvement. Next: RAG-based contract matching (in design).

ManaJet – AI-Assisted Project & Task Management for Small Software Teams **Final Year Project**
Lead System Developer **2022 – 2023**

- Engineered an AI-driven workflow planning system using OpenAI APIs to generate task breakdown structures, timeline estimates, and resource dependencies from user requirements.
- Implemented structured JSON schema validation and prompt constraint rules to ensure deterministic API outputs for backend data persistence.
- Built RESTful API endpoints and relational database schemas (MySQL) to manage dynamic project states and complex task dependency graphs.

OPEN SOURCE

Notemind – Claude Code Plugin for Notes and Reminders github.com/aliyanfaisal/notemind
Author **MIT License**

- Built a free plugin that lets developers add notes in plain words from a Claude Code chat; Claude resolves the date, and notes are saved in a local SQLite database.
- Added a session-start hook (about 85 ms) that reports overdue and upcoming notes, and designed it with least privilege: only the user can add or close notes, never the model.
- Implemented in standard-library Python with no network calls, versioned migrations and a release script.

EXPERIENCE

UrbanSofts Remote
Founder & CEO **2022 – Present**

- Founded UrbanSofts and served as its CEO, leading a remote team of 4–10 members focused on software development; set product direction and oversaw engineering and delivery.

Planning & Development Department, Government of Gilgit-Baltistan Gilgit, Pakistan
Backend Developer & Data Analyst **2024 – 2025**

- Built centralized data analytics dashboards processing public-sector records (XLS, CSV, MySQL) using Python (Pandas) and Laravel backend.
- Digitized administrative paper workflows into structured reporting frameworks to aid decision-support systems for regional planning.

Miami Castle Beach Remote (Part-time)
IT & Web Expert **2023 – 2024**

- Developed and managed WordPress and Flask websites for a Miami Beach castle-rental organization, including server configuration and security.

Greymatter Ventures Pvt Gilgit, Pakistan
Full Stack Web Developer (Team Lead) **2020 – 2022**

- Designed modular backend architectures, relational schemas, and server-side logic to optimize application throughput and API reliability.
- Led the web development team, mentoring junior developers and remote interns on code optimization, system design, and deployment workflows.

Freelance (Fiverr & Upwork) Remote
Applied AI & Backend Engineer **2020 – Present**

- Engineered production-grade applications integrating LLM APIs (OpenAI, Claude, Gemini) for automated document summarization, data extraction, and dynamic e-commerce processing.

LANGUAGES

Urdu: Native · **English:** Advanced

REFERENCES

Dr. Aftab Ahmed Khan aftab.ahmed@kiu.edu.pk
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